

Chain rule # 01

for finding derivative of composite function.

If $z = f(x, y)$ is differentiable fn: of x and y , where $x = g(t)$, $y = h(t)$, are both differentiable fns of t . Then z is a differentiable fn of t .

$$\boxed{\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}}$$

Ex:- If $z = f(x, y) = x^2y + 3xy^4$, where $x = \sin 2t$, $y = \cos t$, find $\frac{dz}{dt}$ at $t = 0$.

Sols:-

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial f}{\partial x} = 2xy + 3y^4$$

$$\begin{aligned} \sin ax &= \\ \cos ax &= a \end{aligned}$$

$$\frac{\partial f}{\partial y} = x^2 + 12xy^3$$

$$\frac{dx}{dt} = \cos 2t \times 2 = 2\cos 2t$$

$$\frac{dy}{dt} = -\sin t \times 1 = -\sin t$$

$$\frac{dz}{dt} = (2xy + 3y^4)(2\cos 2t) + (x^2 + 12xy^3)(-\sin t)$$

Date: _____

Day: M T W T F S S

$$\text{at } \boxed{t=0}, \quad x = \sin 2(t) = \sin 2(0) = \sin 0 = 0, \quad \boxed{x=0}$$
$$y = \cos t = \cos(0) = \cos 0 = 1 \quad \boxed{y=1}$$

$$\frac{dz}{dt} = (0 + 3(1)^4)(2 \cos 2 \times 0) + (0 + 0)(-\sin 0)$$
$$0 = \cos 0 = 1$$
$$2 \times 1$$

$$= (3)(2 \times 1) + 0$$

$$\boxed{\frac{dz}{dt} = 6 \text{ at } t=0}$$

Exp:- $x = f(x, y) = x^3 y^3 + 3x^2 y^2$

$$x = \cos 2t, \quad y = \sin 2t$$

$$\frac{dz}{dt} = ? \quad \text{at } t = 0.$$

Sol:-

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial f}{\partial x} = 3x^2 y^3 + 6xy^2$$

$$\frac{\partial f}{\partial y} = 3x^3 y^2 + 6x^2 y$$

$$\frac{dx}{dt} = -\sin 2t \times 2 = -2 \sin 2t$$

$$\frac{dy}{dt} = \cos 2t \times 2 = 2 \cos 2t$$

Date: _____

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$$\frac{dz}{dt} = (3x^2y^3 + 6xy^2)(-2\sin 2t) + (3x^3y^2 + 6x^2y)(2\cos 2t)$$

$$\text{at } t=0, x = \cos 2t = \cos 2(0) = \cos 0 = 1, \boxed{x=1}$$

$$y = \sin 2t = \sin 2(0) = \sin 0 = 0, \boxed{y=0}$$

~~$$(3(1)^2(0)^3 + 6(1)(0)^2)(-2\sin 2 \times 0) + (3(1)^3(0)^2 + 6(1)^2(0))(2\cos 2 \times 0)$$~~

$$\frac{dz}{dt} = (3(1)^2(0)^3 + 6(1)(0)^2)(-2\sin 2 \times 0) + (3(1)^3(0)^2 + 6(1)^2(0))(2\cos 2 \times 0)$$

$$= (0 + 0)(0) + (0 + 0)(2 \times 1)$$

$$0 = \cos 0 = 1$$

$$2 \times 1 = 2$$

$\frac{dz}{dt} = 0 \text{ at } t=0$
